

This is the complete data pipeline and architectural specification for the MetroFlow AI platform, designed strictly according to the provided Product Requirements Document (PRD) and Research Report.

1. DATA REQUIREMENTS

Based on the uploaded documents, here is the exact data required to feed the analytical engine.

A. Traffic Count Data

- **Field Name:** `Timestamp` | **Type:** Datetime | **Unit:** YYYY-MM-DD HH:MM:SS | **Example:** 2026-08-08 07:45:00 | **Mandatory:** Yes | **Why:** Temporal cluster analysis for peak windows. | **Source/Support:** *Requirement from PRD* (hourly/minute counts). PDF+ 1
- **Field Name:** `Intersection_ID` | **Type:** String | **Unit:** None | **Example:** INT_04 | **Mandatory:** Yes | **Why:** Distinguishes multi-intersection corridor data. | **Source/Support:** *Requirement from PRD* (multi-intersection scenario). PDF+ 1
- **Field Name:** `Approach_Direction` | **Type:** String | **Unit:** None | **Example:** Eastbound | **Mandatory:** Yes | **Why:** To map volumes to specific signal phases. | **Source/Support:** *Requirement from PRD* (approach-level analysis). PDF+ 1
- **Field Name:** `Volume` | **Type:** Integer | **Unit:** Vehicles/interval | **Example:** 145 | **Mandatory:** Yes | **Why:** Core metric for volume-to-capacity calculations. | **Source/Support:** *Finding from Research & PRD* (traffic counts). PDF+ 2

B. Road Metadata

- **Field Name:** `Intersection_ID` | **Type:** String | **Unit:** None | **Example:** INT_04 | **Mandatory:** Yes | **Why:** Relational key to traffic counts. | **Source/Support:** *Requirement from PRD*. PDF+ 1
- **Field Name:** `Approach_Direction` | **Type:** String | **Unit:** None | **Example:** Eastbound | **Mandatory:** Yes | **Why:** Defines the geometry of the specific leg. | **Source/Support:** *Requirement from PRD*. PDF+ 1
- **Field Name:** `Lanes` | **Type:** Integer | **Unit:** Count | **Example:** 3 | **Mandatory:** Yes | **Why:** Calculates total approach capacity. | **Source/Support:** *Requirement from PRD*. PDF+ 1
- **Field Name:** `Speed_Limit` | **Type:** Integer | **Unit:** mph/kmh | **Example:** 45 | **Mandatory:** Yes | **Why:** Needed for yellow/red clearance safety bounds. | **Source/Support:** *Requirement from PRD*. PDF+ 1

- **Field Name:** *Storage_Bay_Length* | **Type:** Float | **Unit:** Meters | **Example:** 50.0 | **Mandatory:** No | **Why:** Queue spillback risk scoring on short turn lanes. | **Source/Support:** *Requirement from PRD (US-11).*
PDF+ 1

C. Turning Movement Data

- **Field Name:** *Intersection_ID* | **Type:** String | **Unit:** None | **Example:** INT_04 | **Mandatory:** Yes | **Why:** Relational key. | **Source/Support:** *Requirement from PRD.*
PDF+ 1
- **Field Name:** *Approach_Direction* | **Type:** String | **Unit:** None | **Example:** Eastbound | **Mandatory:** Yes | **Why:** Defines which approach the ratios apply to. | **Source/Support:** *Requirement from PRD.*
PDF+ 1
- **Field Name:** *Left_Ratio* | **Type:** Float | **Unit:** Decimal (0-1) | **Example:** 0.15 | **Mandatory:** Yes | **Why:** Splits approach volume into distinct phase demands. | **Source/Support:** *Requirement from PRD (US-03).*
PDF+ 1
- **Field Name:** *Through_Ratio* | **Type:** Float | **Unit:** Decimal (0-1) | **Example:** 0.75 | **Mandatory:** Yes | **Why:** Calculates through-movement saturation. | **Source/Support:** *Requirement from PRD (US-03).*
PDF+ 1
- **Field Name:** *Right_Ratio* | **Type:** Float | **Unit:** Decimal (0-1) | **Example:** 0.10 | **Mandatory:** Yes | **Why:** Completes total approach demand breakdown. | **Source/Support:** *Requirement from PRD (US-03).*
PDF+ 1

D. Signal Timing Data (Baseline)

- **Field Name:** *Intersection_ID* | **Type:** String | **Unit:** None | **Example:** INT_04 | **Mandatory:** Yes | **Why:** Relational key. | **Source/Support:** *Requirement from PRD.*
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- **Field Name:** *Phase_ID* | **Type:** Integer | **Unit:** None | **Example:** 2 | **Mandatory:** Yes | **Why:** Maps to NEMA standard phases. | **Source/Support:** *Requirement from PRD.*
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- **Field Name:** *Current_Green* | **Type:** Integer | **Unit:** Seconds | **Example:** 45 | **Mandatory:** Yes | **Why:** Baseline for before vs. after simulation. | **Source/Support:** *Requirement from PRD.*
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- **Field Name:** *Min_Ped_Walk* | **Type:** Integer | **Unit:** Seconds | **Example:** 18 | **Mandatory:** Yes | **Why:** Hard safety constraint for optimization. | **Source/Support:** *Requirement from PRD.*
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- **Field Name:** *Yellow_Time* | **Type:** Float | **Unit:** Seconds | **Example:** 4.0 | **Mandatory:** Yes | **Why:** Statutory safety bound enforcement. | **Source/Support:** *Requirement from PRD (FR-C04).*
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E. Incident Data

- **Field Name:** `Incident_ID` | **Type:** String | **Unit:** None | **Example:** INC_001 | **Mandatory:** No | **Why:** Tracks unique events. | **Source/Support:** *Requirement from PRD.*
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- **Field Name:** `Timestamp_Start` | **Type:** Datetime | **Unit:** YYYY-MM-DD HH:MM:SS | **Example:** 2026-08-08 14:00:00 | **Mandatory:** Conditional (if uploaded) | **Why:** Filters out anomalies from baseline timing. | **Source/Support:** *Requirement from PRD (US-04).*
PDF+ 1
- **Field Name:** `Event_Type` | **Type:** String | **Unit:** None | **Example:** Roadwork | **Mandatory:** Conditional | **Why:** Explains volume drops or delay spikes. | **Source/Support:** *Requirement from PRD (US-04).*
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2. SYNTHETIC DATA STRATEGY

To successfully demonstrate the MVP, the synthetic data must replicate real-world traffic physics and daily commuter rhythms.

- **Dataset Structure:** A 5-intersection continuous arterial corridor (Main Street) cross-sectioned by minor side streets.
- **Number of Approaches:** 4 approaches per intersection (N/S/E/W).
- **Time Resolution:** 15-minute intervals.
- **Duration:** 7 continuous days (Monday - Sunday).
- **Expected Congestion Pattern:**
 - *AM Peak (07:45 - 09:15):* Heavy Eastbound directional flow (commuters heading to city center).
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 - *Midday (11:30 - 13:00):* Balanced, moderate flow.
 - *PM Peak (16:30 - 18:00):* Heavy Westbound directional flow (commuters heading home).
 - *Off-Peak:* Low, stable volume.
- **Expected Bottleneck:**
 - *Requirement from PRD:* Intersection #4 (Main St & 5th Ave) must be the primary bottleneck with a V/C ratio of 1.12 (LOS F).
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- **Generation Strategy [Recommended Enhancement for Hackathon]:** Use a deterministic sine-wave function layered with random Gaussian noise to generate base daily traffic. Multiply Eastbound AM intervals by a factor of 2.5 and Westbound PM intervals by 2.5. Cap the physical capacity (lanes × saturation flow) so that Intersection #4 mathematically triggers $V/C > 1.0$ and initiates queue spillback in the simulation.

3. DATA VALIDATION

Requirement from PRD: Detect and flag corrupted, missing, or outlier data points with options for interpolation or exclusion (FR-A03).

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Problem	Detection Method	Action	Warning/Error Message	Continue?
Missing Values	Null checks on Volume or Timestamp .	Interpolate using adjacent 15-min intervals.	<i>Warning: Missing volumes interpolated for [Time].</i>	Yes
Duplicate Records	Group by ID + Timestamp.	Keep latest record, discard duplicates.	<i>Warning: Duplicate timestamps removed.</i>	Yes
Invalid Timestamps	Regex format validation.	Reject row if unparseable.	<i>Error: Row 45 timestamp format invalid.</i>	No (if > 5% missing)
Negative Counts	Volume < 0.	Set to 0.	<i>Warning: Negative volumes floored to 0.</i>	Yes
Unrealistic Volumes	Volume > (Lanes * 2200 veh/hr).	Cap at maximum theoretical saturation flow.	<i>Warning: Implausible volume at INT_4. Capped at saturation limit.</i>	Yes
Invalid Turning Sum	Left + Through + Right != 1.0.	Normalize ratios to sum to 1.0.	<i>Warning: Turn ratios at INT_2 normalized.</i>	Yes
Missing Speed Limits	Null check on metadata.	Default to 35 mph.	<i>Warning: Missing speed limit. Defaulting to 35 mph.</i>	Yes

4. DATA PIPELINE

Requirement from PRD: The end-to-end architecture (Module A to Module E).

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1. **USER UPLOAD:** Traffic engineer uploads historical 15-minute volume logs and road geometry via CSV to the web portal.

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2. **DATA VALIDATION:** System parses counts, checks for invalid/missing data, and applies sanitization rules.

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3. **DATA CLEANING:** Outliers (e.g., equipment malfunctions showing 10,000 veh/hr) are clamped to physical maximums.
4. **DATA TRANSFORMATION:** Volumes are aggregated into hourly flow rates to match Highway Capacity Manual (HCM) standards.
5. **TRAFFIC ANALYSIS:** System calculates base saturation flows based on lane geometries.
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6. **PEAK DETECTION:** Temporal cluster analysis scans the 7-day dataset to identify the highest sustained 1-hour volume blocks.
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7. **BOTTLENECK DETECTION:** Computes the Volume-to-Capacity (V/C) ratio. Flags any intersection >1.0 as a bottleneck.
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8. **SIGNAL OPTIMIZATION:** Deterministic Python engine applies Webster's method to compute optimal cycle length and allocates green time proportional to critical lane volumes.
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9. **SAFETY VALIDATOR:** Checks new splits against minimum pedestrian walk times and clearance intervals. Rejects/modifies if failed.
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10. **SIMULATION:** M/M/1 queueing formulas estimate "before vs after" delay and queue lengths.
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11. **AI EXPLANATION:** Structured JSON of the before/after metrics is sent to Azure OpenAI to generate a plain-text rationale.
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12. **DASHBOARD & EXPORT:** Metrics, UI widgets, and PDF/CSV downloads are rendered for the user.
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5. PEAK CONGESTION ANALYSIS

Requirement from PRD: Identify peak congestion windows automatically using temporal cluster analysis (FR-B02).

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- **Inputs:** Cleaned 15-minute interval total intersection volumes.
- **Calculation:**
 1. Apply a rolling 1-hour sum window across the 24-hour timeline.
 2. Group by time-of-day across the 7-day dataset to find the average hourly volume for each 15-minute start time.
 3. The absolute maximum rolling sum identifies the **AM Peak**, **Midday Peak**, and **PM Peak** boundaries.
- **Output:** Time windows (e.g., 07:45 - 08:45 AM).
- **Visualization:** An hourly volume bar chart with shaded overlay boxes highlighting the identified peak periods.
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- **Edge Cases:** Flat volume profiles (e.g., weekends) where peaks are not distinct. The system will default to the highest continuous 60-minute segment.

6. BOTTLENECK DETECTION

Requirement from PRD & Research: Calculate V/C ratios, assign HCM LOS based on delay, and map multi-intersection bottlenecks.

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- **Capacity (c):** Number of Lanes × Saturation Flow (default 1,900 veh/hr/lane under ideal conditions).
- **Traffic Demand (v):** Hourly flow rate derived from the peak 15-minute volume.
- **V/C Ratio (X):** Calculated as v/c .
- **Worked Synthetic Example:**
 - *Intersection 4, Eastbound Approach.*
 - Demand (v): 2,120 veh/hr.
 - Lanes: 2. Saturation Flow: 1,900 veh/hr/lane.
 - Green Ratio (g/C): 0.45.
 - Capacity (c): $2 \times 1,900 \times 0.45 = 1,710$ veh/hr.
 - **V/C Ratio:** $2,120 / 1,710 = 1.24$.
- **Ranking:** Intersections are ranked descending by their highest approach V/C ratio. V/C > 1.0 is flagged as a bottleneck (LOS F).

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7. SIGNAL TIMING RECOMMENDATION

Requirement from PRD & Research: Use Webster's method to calculate cycle lengths and splits offline.

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- **Algorithm Step-by-Step (Deterministic Python Module):**
 1. **Traffic Demand:** Identify critical lane volumes (vc) for conflicting phases (e.g., North/South vs East/West).
 2. **Saturation Ratio (y):** Calculate $y = vc/s$ for each critical phase. Sum these to find intersection total $Y = \sum y$.
 3. **Lost Time (L):** Calculate total lost time per cycle (clearance intervals).
 4. **Cycle Length (C0):** Apply Webster's Formula:
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 $C0 = 1 - Y \cdot 1.5L + 5$
 5. **Cycle Bounds:** Constrain C0 to user-defined limits (e.g., 60s min, 180s max) (FR-C02).
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 6. **Green Split:** Distribute effective green time proportionally to critical ratios:
 $gi = (C0 - L) \times Y_{yi}$

7. **Corridor Offset:** Calculate offsets based on distance and speed limit to create a green wave (FR-C03).

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- **Assumptions [Must be explicitly displayed]:** Steady-state arrival patterns; default saturation flow of 1,900 veh/hr/lane; no heavy vehicle/grade impedance unless manually specified.

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8. SAFETY GUARDRAILS

Requirement from PRD: Hard safety lower-bounds for minimum green, yellow change, red clearance, and pedestrian walk times (FR-C04).

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- **Pedestrian Walk Time:** Calculated based on crosswalk distance divided by walk speed (e.g., 3.5 ft/sec).
- **Yellow Interval:** Calculated using approach speed limit and standard deceleration rates.
- **Validation Layer:** Before the Python optimization engine outputs its result, the `Safety_Validator` module intercepts the array.
 - *Check:* `If Calculated_Green[Phase_2] < Min_Ped_Walk[Phase_2]`
 - *Action:* Override `Calculated_Green` to equal `Min_Ped_Walk`.
Re-distribute the remaining cycle time to other phases.
 - If mathematical resolution is impossible without exceeding max cycle length, the system rejects the calculation and raises a flag: "Optimization blocked: Volume demands conflict with mandatory pedestrian safety minimums."

9. BEFORE VS AFTER SIMULATION

Requirement from PRD: Simulate queue accumulation/dissipation using M/M/1 and Webster queuing formulation logic, estimate delay, fuel, and CO₂, and explicitly label as estimates.

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- **Model Used:** Webster's Delay Formula & M/M/1 Queuing.
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- **Calculate Uniform Delay (d1):**

$$d1 = 1 - [\min(1, X) \times (g/C)] / 0.5C(1 - g/C)^2$$
- **Calculate Random Delay (d2):** Assessed using M/M/1 queuing approximations.
- **Queue Length:** $Q = \text{Arrival Rate} \times \text{Delay}$.
- **Display Logic:**
 - Current Signal Plan → Calculates Delay X, Queue Y, LOS Z.
 - Recommended Signal Plan → Calculates Delay A, Queue B, LOS C.
 - Delta metrics: "Average intersection delay reduced by 35% (SIMULATED ESTIMATE)."

- **Emissions [Requirement from PRD]:** Calculate fuel/CO2 based on idling time (e.g., 0.5 gallons per hour of vehicle idle delay).
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- **Limitations Disclaimer:** "This simulation assumes stable flow. Actual field results will vary. No real-world guaranteed improvements are claimed."
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10. WHAT-IF STRESS TESTING

Requirement from PRD: Interactive slider controls allowing users to perform traffic volume variation stress testing (+/- 30%).

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- **Design:** A web UI slider on the Simulation screen mapping to multipliers: [0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3].
- **Logic:** When the user moves the slider to "+20%", the frontend passes a 1.2 demand multiplier to the Python API. The backend recalculates the V/C ratio, Delay, and Queue length using the new demand over the fixed, recommended capacity.
- **Output:** The UI updates dynamically, showing if the proposed plan fails (LOS drops to F) under a 20% surge, proving/disproving plan robustness.
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11. INCIDENT HANDLING

Requirement from PRD: Optional ingestion of incident logs (accidents, construction) tagged by timestamp and location.

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- **Representation:** CSV upload of Incident_ID, Location, Start_Time, End_Time.
- **Logic:** The data parsing engine checks traffic counts against incident time-windows.
- **Action:** If a major volume drop or delay spike correlates with a known accident window, the system automatically excises that block of data from the "baseline" profile used to calculate green splits, preventing an abnormal day from skewing a 5-year fixed timing plan.
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12. AI ROLE

Requirement from PRD: Azure OpenAI must provide plain-language explanations and convert math to narrative, but MUST NOT calculate math or override safety.

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- **Architecture Flow:** DETERMINISTIC TRAFFIC ENGINE → CALCULATED RESULTS → STRUCTURED JSON → AZURE OPENAI → PLAIN-LANGUAGE EXPLANATION
 - **Prompt Engineering (Internal):** "You are an expert traffic engineer advisor. You are provided with a structured JSON containing a before-and-after signal timing change and the resulting simulated delay. Explain to a city council why Phase 2 green time was increased by 14 seconds based strictly on the JSON data. Do not invent traffic data or make guaranteed improvement claims."
 - **Example Expected Output:** "Green time for Phase 2 (Eastbound Through) was increased by 14 seconds to accommodate the morning rush surge observed between 08:00 AM - 08:45 AM. Minimum pedestrian clearance times (Phase 4/8) were strictly preserved at 18 seconds."
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13. DASHBOARD DATA

Requirement from PRD: Map data needs for the screen-by-screen user journey.

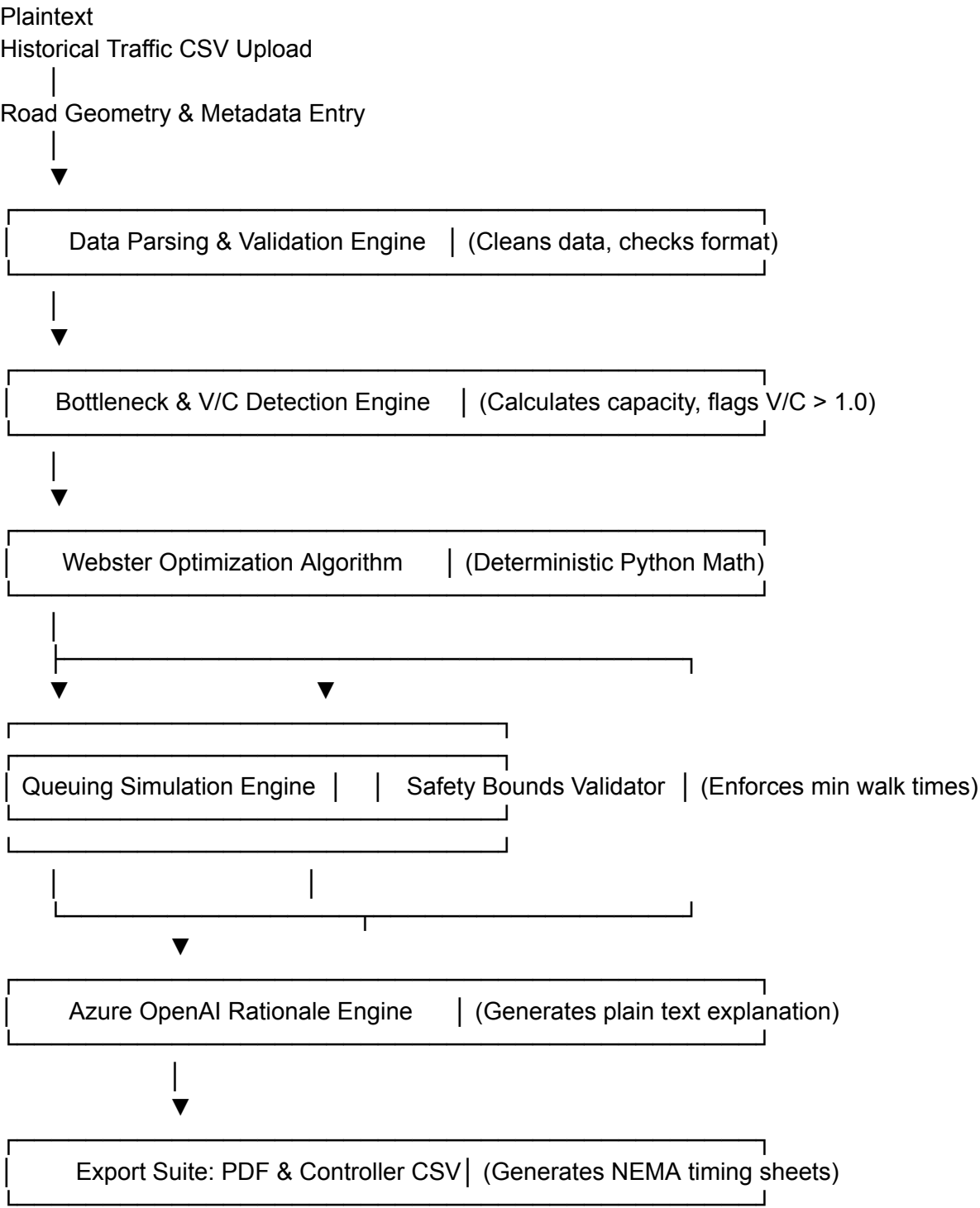
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Dashboard Screen	Input/Calculated Data	Visuals/Charts	User Actions
Data Ingestion	Raw CSVs, Intersection Geometry, Incident Logs.	File upload progress bar, Interactive node map.	Upload files, map approach lanes, set speed limits.
Congestion Analyzer	Rolling 1-hr sums, V/C Ratios, LOS per approach.	Corridor Heatmap, Hourly volume bar charts.	View peak hours, identify bottlenecks.
Signal Optimization	Recommended Cycle Length, Phase Splits, Offsets.	Phase diagram, minimum safety threshold alerts.	Adjust bounds, lock preemptions, trigger optimizer.
Simulation	M/M/1 Queue outputs, Baseline vs Recommended Delay.	Side-by-side metric cards (Delay, Queue, LOS).	Adjust "What-if" slider (+/- 30%), view CO2 impact.
AI Rationale & Export	JSON payload from Python engine, Azure OpenAI text.	Plain-text Markdown panel, Controller CSV preview.	Download PDF Report, Download NEMA CSV.

[Recommended Enhancement for Hackathon MVP]: Integrate an interactive Time-Space progression diagram on the Signal Optimization screen to visualize the corridor green wave, as heavily recommended by the PRD's self-evaluation section.

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14. FINAL DATA ARCHITECTURE



Requirement from PRD: Architecture based on Feature Dependency Map.

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15. MVP VS OPTIONAL

Requirement from PRD: Feature prioritization based on the MoSCoW matrix.

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MUST BUILD FOR HACKATHON (MVP Priority):

- CSV Upload & Basic Metadata Entry.
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- Peak Window ID & V/C Ratio calculation.
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- Webster Green Splits & Cycle Bounds engine.
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- Queuing Theory Delay & Queue Simulation (M/M/1).
PDF
- Volume Surge Stress Test Slider.
PDF
- LLM Rationale Generation & PDF/CSV Export.
PDF
- Interactive Time-Space Progression Diagrams (added as priority based on PRD review section).
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OPTIONAL ENHANCEMENTS (Post-MVP):

- Batch Ingestion & GIS Shapefile Auto-import.
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- Weather Impact Correlation.
PDF
- Day-of-Week Schedule Banding.
PDF
- Custom Municipal Branding for PDFs.
PDF
- Deep Reinforcement Learning (SUMO) integration.
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16. FINAL IMPLEMENTATION SPECIFICATION

For the developer building this platform:

1. **"WHAT DATA DO WE NEED?"** You need a continuous 7-day CSV of 15-minute traffic counts, tagged by intersection and approach, alongside basic road geometry (lanes, speed limits). Generate this synthetically using a sine-wave baseline heavily weighted in the Eastbound AM and Westbound PM directions.
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2. **"WHAT DO WE DO WITH THAT DATA?"** Validate it (remove negatives/nulls), aggregate it into hourly flow rates, and calculate the V/C ratio by dividing the volume by the approach capacity (lanes × saturation flow).
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3. **"HOW DO WE CALCULATE THE RESULT?"** Use deterministic Python to run Webster's formula $C0 = (1.5L + 5) / (1 - Y)$. Distribute the resulting green time based on the ratio of competing critical lane volumes. Pass this through a safety check that

hard-stops any green time below the pedestrian minimum walk time.

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4. **"HOW DOES THE AI USE THE RESULT?"** Do not let the AI do math. Take the before-and-after simulation numbers generated by your Python queuing equations, wrap them in a JSON, and prompt Azure OpenAI to write a 2-paragraph summary explaining *why* the splits changed.

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5. **"WHAT DOES THE USER SEE?"** A clean web dashboard showing a corridor heatmap, side-by-side delay/queue reduction metrics, a "what-if" volume slider, and downloadable NEMA CSV / PDF reports. Every screen must clearly label outputs as offline simulation estimates.

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